SYNCHRONOUS DIGITAL HIERARCHY (SDH)

EVOLUTION TEKNOLOGI SDH

❖ Sejarah SDH

1. **1970 – pengenalan teknik PCM dalam jaringan telekomunikasi**

2. **Teknik multiplex sinkron dengan menggunakan PCM 32 slot (PCM 30) dengan bit rate 2.048Mbit/s (E1) di Eropa**

3. **Teknik Multiplex kecepatan yang lebih tinggi melalui PDH, menggunakan pulsa-pulsa *clock y*ang sangat akurat di seluruh simpul penyakelarnya (*switching node*) .**

4. **1985 - Perusahaan Bellcore memperkenalkan SONET (Synchronous Optical Network), karena ditemukan kekurangan PDH (Hirarki Transmisi PCM), yang tidak bisa mengatasi penambahan kapasitas yang besar dan keseragaman standar (yang sebelumnya ada 3 standar, ada standar PCM 30/ Eropa, Standar PCM 24/ Amerika, Standar Jepang).**

5. **1988 - konsep SONET diadopsi oleh ITU (International Telecommunication Union) dan ETSI (European Telecommunication Standards Institute) dan diberi nama SDH (Synchronous Digital Hierarchy).**

❖ PLESIOCHRONOUS DIGITAL HIERARCHY (PDH)

Plesiochronous Multiplexing

- Before SDH transmission networks were based on the PDH hierarchy.
- Plesiochronous means nearly synchronous.
- 2 Mbit/s service signals are multiplexed to 140 Mbit/s for transmission over optical fiber or radio.
- Multiplexing of 2 Mbit/s to 140 Mbit/s requires two intermediate multiplexing stages of 8 Mbit/s and 34 Mbit/s.
- Multiplexing of 2 Mbit/s to 140 Mbit/s requires multiplex equipment known as 2, 3 and 4 DME.
- Alarm and performance management requires separate equipment in PDH.

❖ PLESIOCHRONOUS DIGITAL HIERARCHY (PDH)

➢ PULSE CODE MODULATION

- Voice Frequency ranges upto 4 Khz
- Sampling the Voice Signal @ 8 Khz (Double the Max. Frequency)
- 8 bits per sample
- Digital Bit Rate: 8Khz X 8 = 64 Kbps

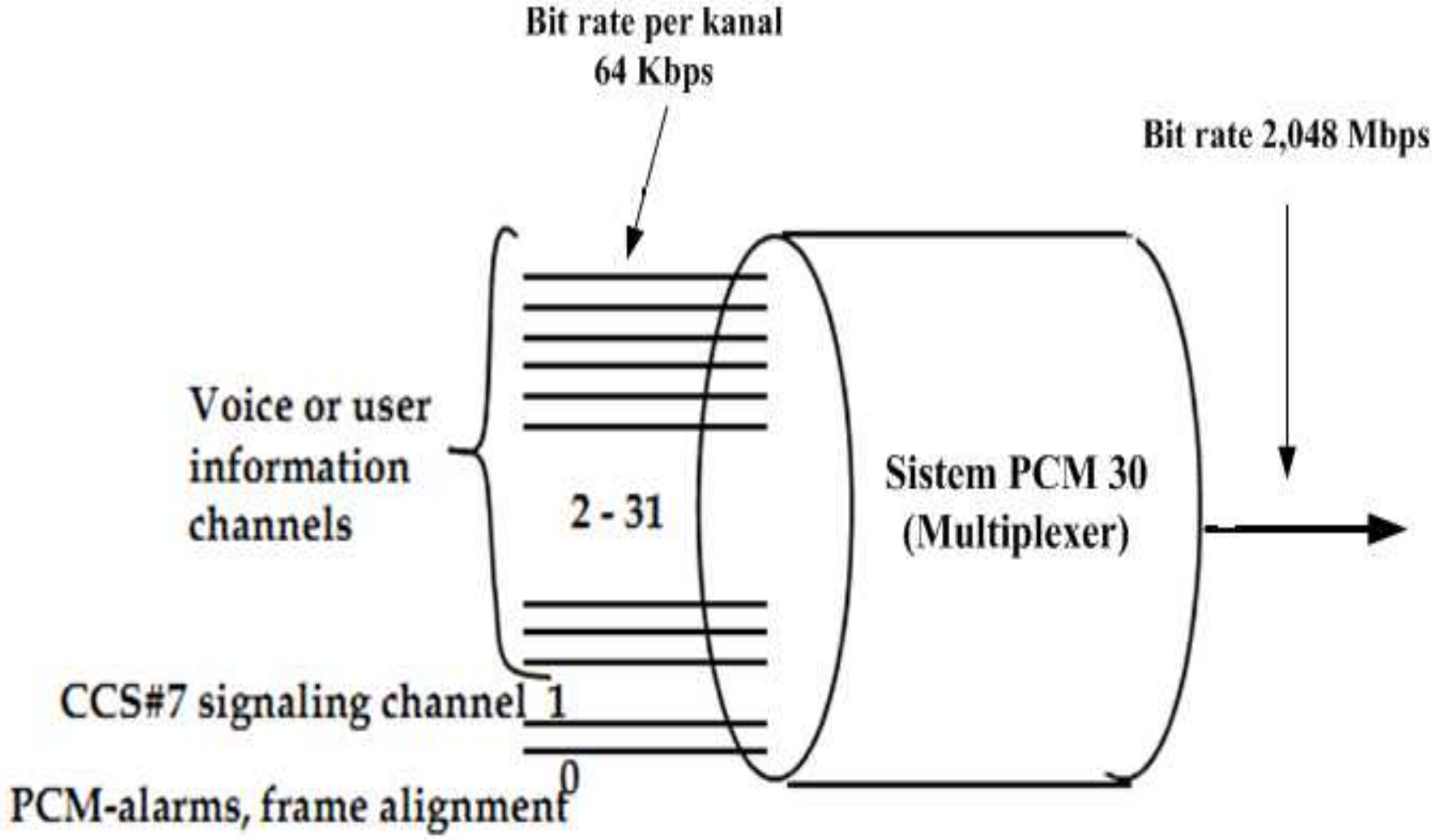
➢ BUILDING UP THE BASE STREAM (2MB)

- 30 Voice Channels @ 64 Kbps
- One channel for Frame (64 K)
- One channel for Signaling (64 K)
- Total number of Channels = 32
- Bit Rate: 32 X 64 K= 2048 Khz (2Mb)

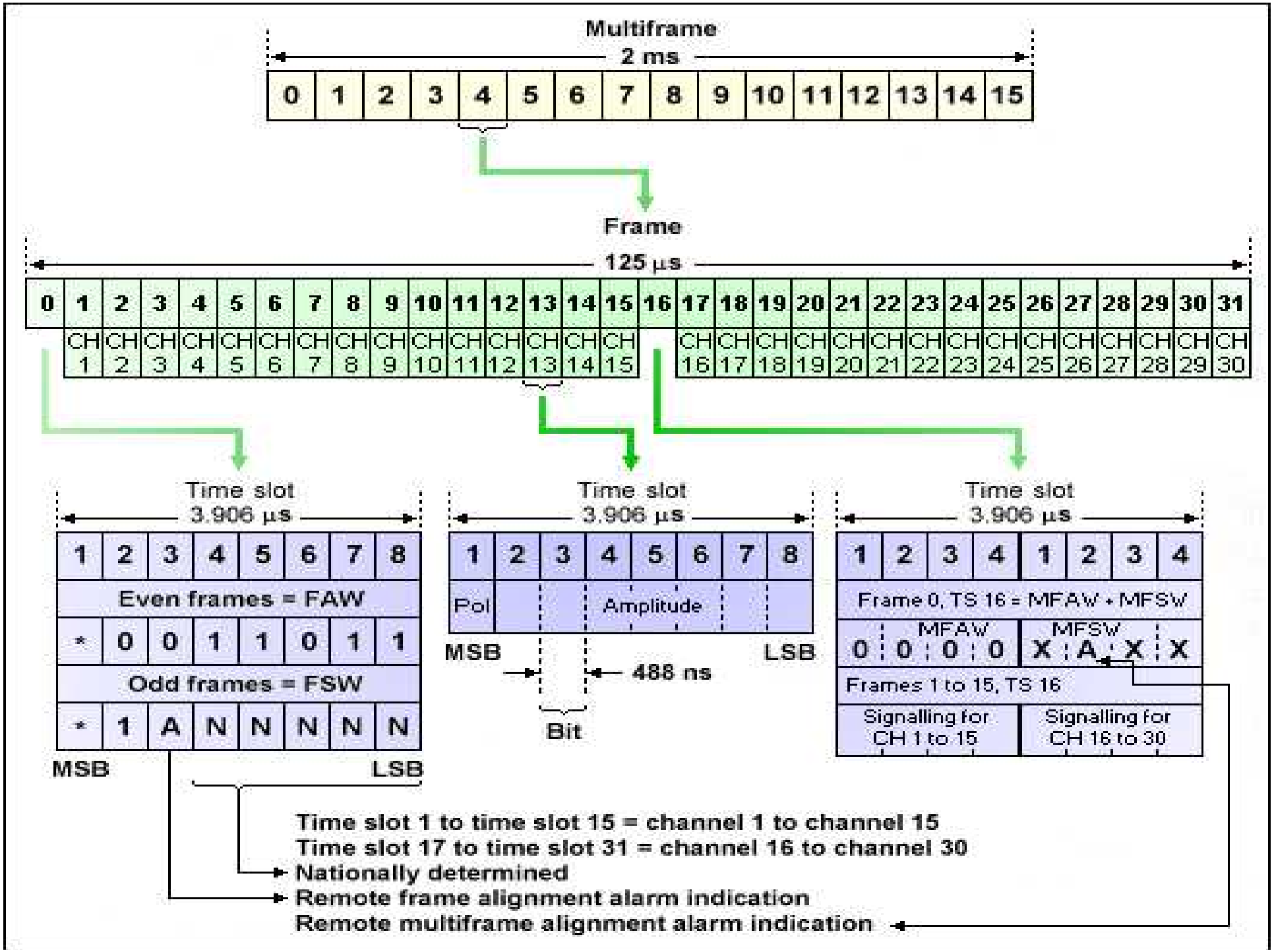
➢ PDH BIT RATES

- E1-2048 Kbps (2Mb) [30 Voice Channel]
- E2-8448 Kbps (8Mb) [120 Voice Channel]
- E3-34368 Kbps (34Mb) [480 Voice Channel]
- E4-139264 Kbps (140Mb) [1920 Voice Channel]

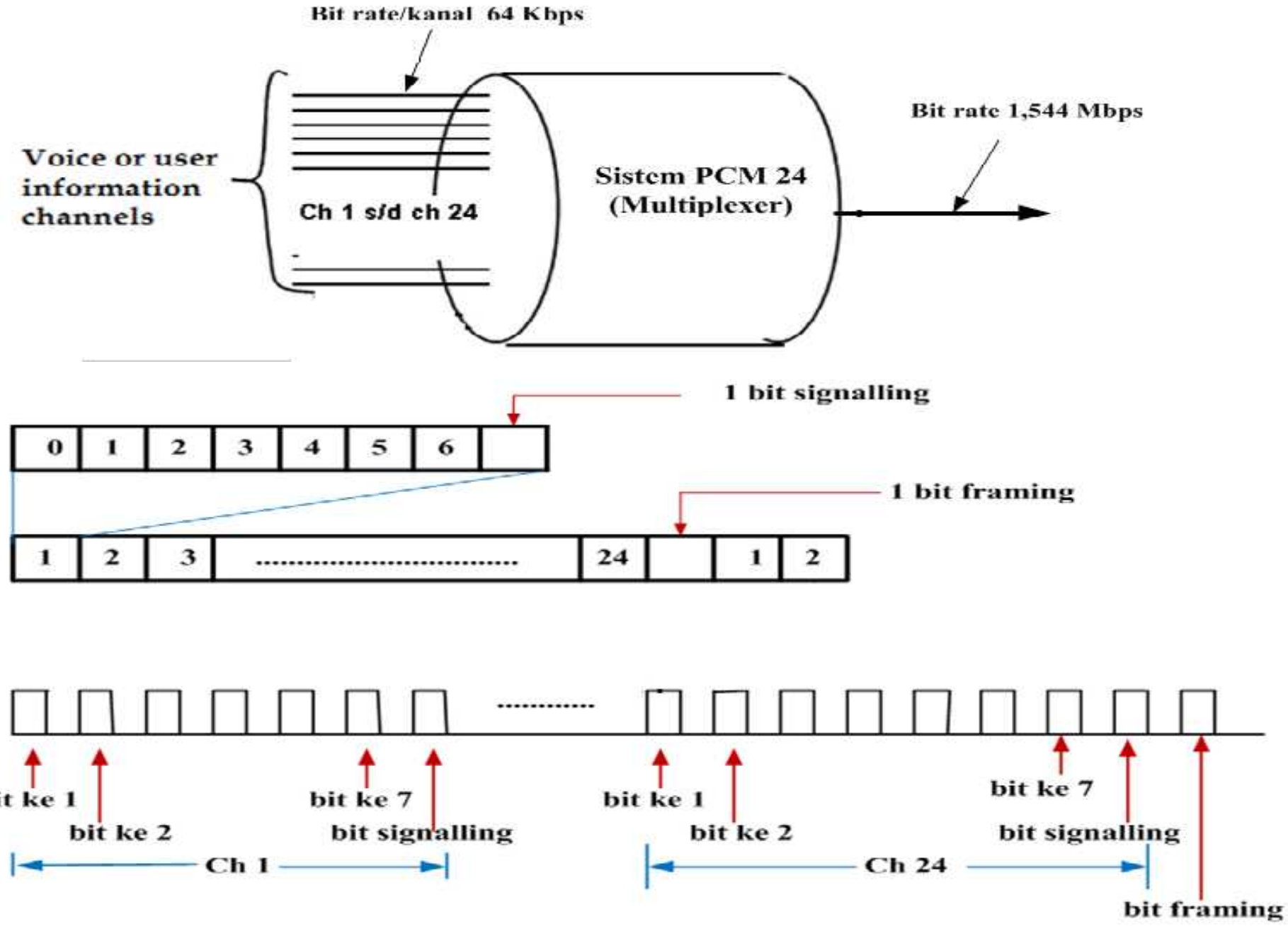
➢ Susunan Frame PCM 30



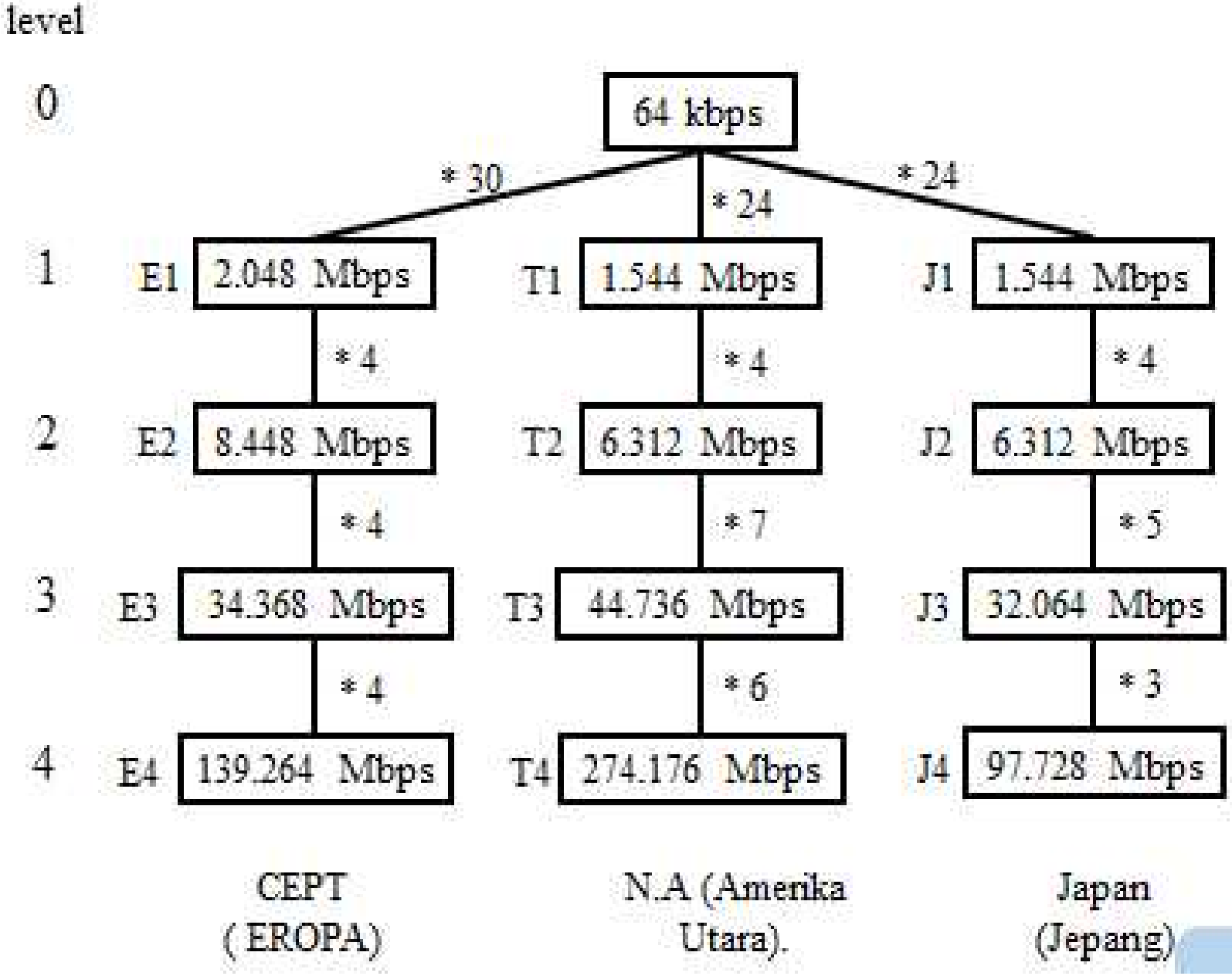
TS no 0 untuk framing
TS no 1 s/d no TS 15 untuk voice
TS no 17 s/d TS no 31 untuk voice
TS no 16 untuk signalling

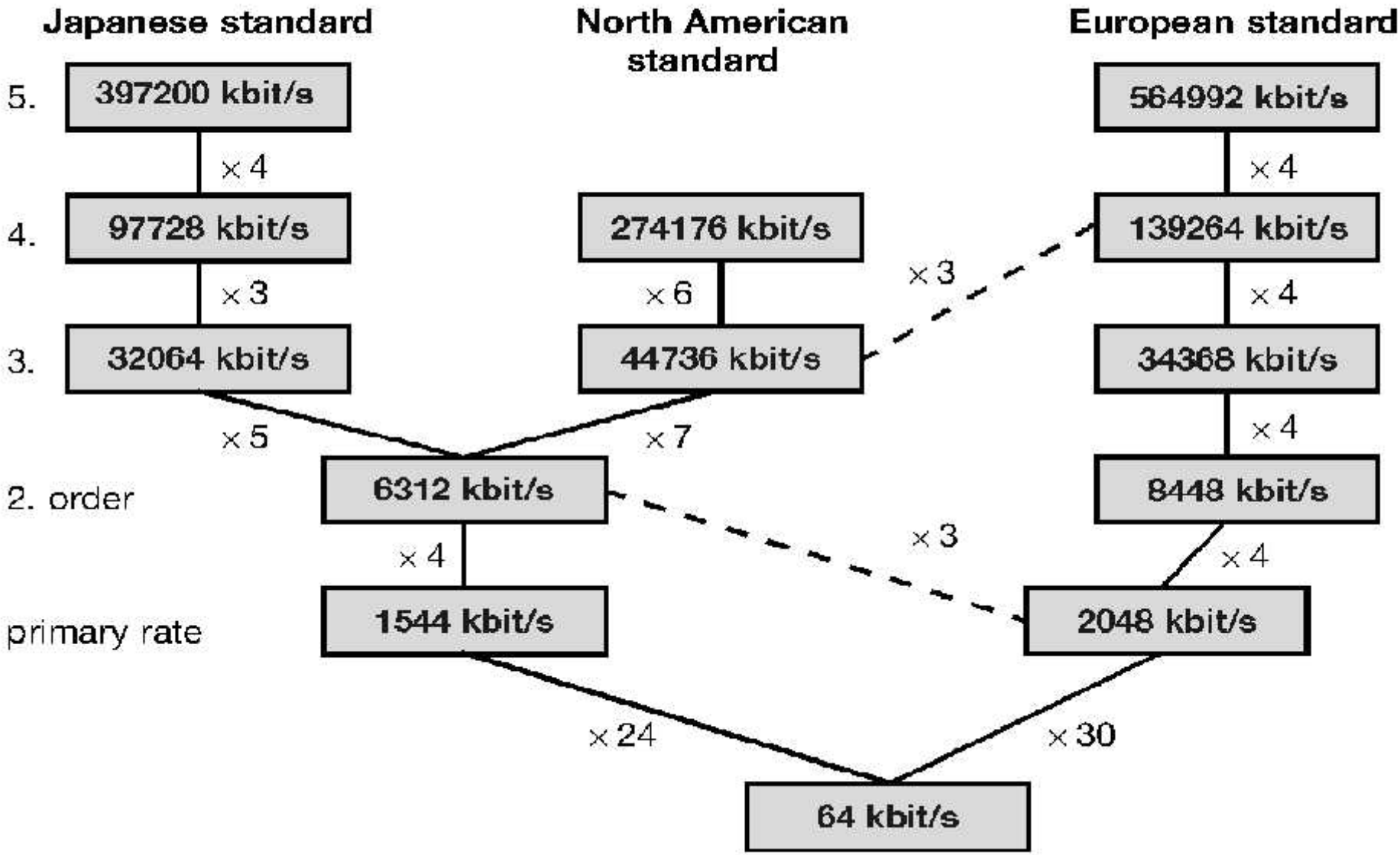

Multiframe
2 ms
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15
Frame
125 µs
0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31
CH 1 CH 2 CH 3 CH 4 CH 5 CH 6 CH 7 CH 8 CH 9 CH 10 CH 11 CH 12 CH 13 CH 14 CH 15
CH 16 CH 17 CH 18 CH 19 CH 20 CH 21 CH 22 CH 23 CH 24 CH 25 CH 26 CH 27 CH 28 CH 29 CH 30
Time slot
3.906 µs
1 2 3 4 5 6 7 8
Even frames = FAW
* 0 0 1 1 0 1 1
Odd frames = FSW
* 1 A N N N N N
MSB
LSB
Time slot
3.906 µs
1 2 3 4 5 6 7 8
Pol
Amplitude
MSB
LSB
488 ns
Bit
Time slot
3.906 µs
1 2 3 4 1 2 3 4
Frame 0, TS 16 = MFAW + MFSW
MFAW
0 0 0 0
MFSW
X A X X
Frames 1 to 15, TS 16
Signalling for CH 1 to 15
Signalling for CH 16 to 30
Time slot 1 to time slot 15 = channel 1 to channel 15
Time slot 17 to time slot 31 = channel 16 to channel 30
Nationally determined
Remote frame alignment alarm indication
Remote multiframe alignment alarm indication

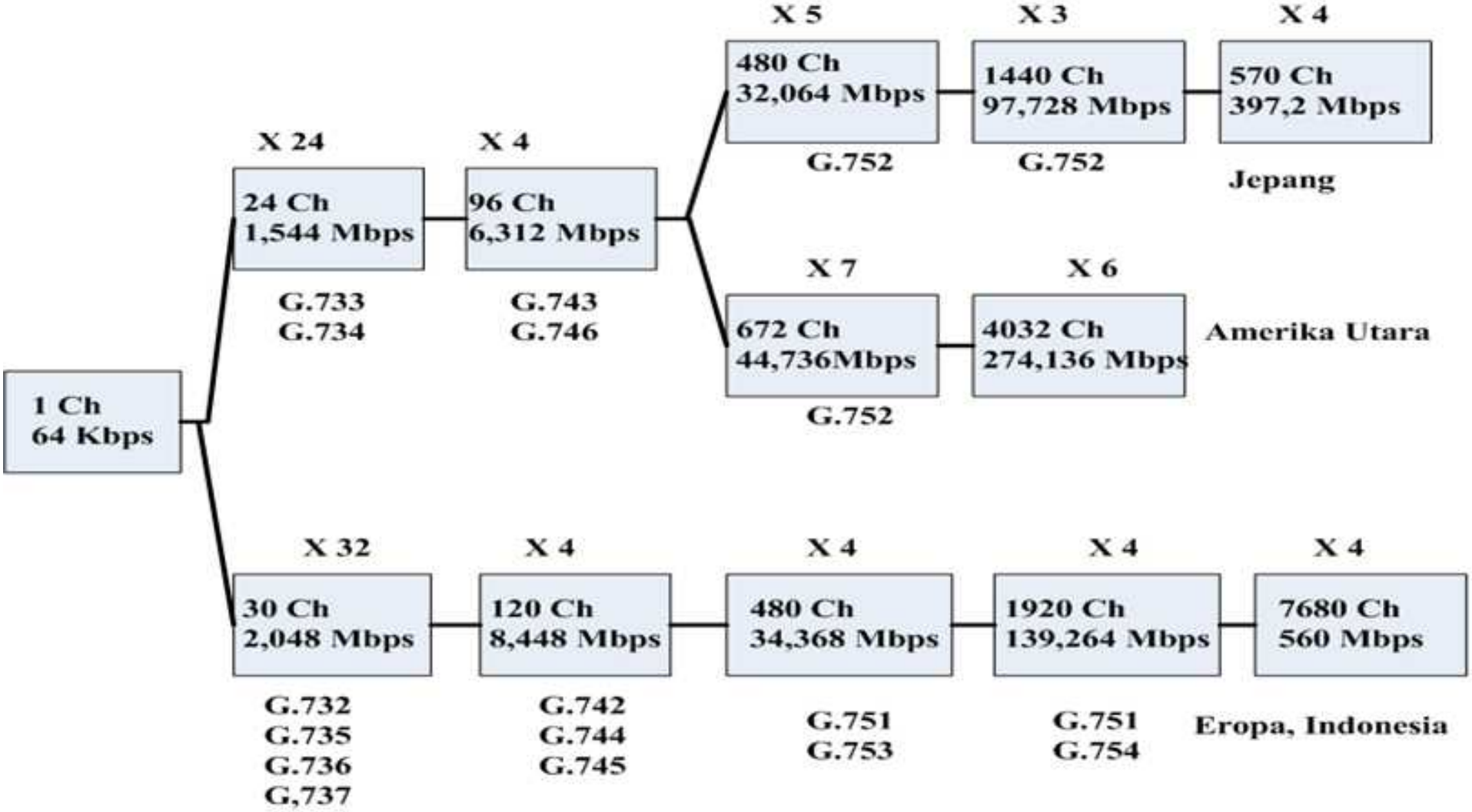
➢ Susunan Frame PCM 24



➢ PDH hierarchies




Japanese standard
North American standard
European standard
5.
397200 kbit/s
564992 kbit/s
×4
×4
4.
97728 kbit/s
274176 kbit/s
139264 kbit/s
×3
×6
×3
×4
3.
32064 kbit/s
44736 kbit/s
34368 kbit/s
×5
×7
×4
2. order
6312 kbit/s
8448 kbit/s
×3
×4
×4
primary rate
1544 kbit/s
2048 kbit/s
×24
×30
64 kbit/s


1 Ch
64 Kbps
X 24
24 Ch
1,544 Mbps
G.733
G.734
X 4
96 Ch
6,312 Mbps
G.743
G.746
X 5
480 Ch
32,064 Mbps
G.752
X 3
1440 Ch
97,728 Mbps
G.752
X 4
570 Ch
397,2 Mbps
Jepang
X 7
672 Ch
44,736Mbps
G.752
X 6
4032 Ch
274,136 Mbps
Amerika Utara
X 32
30 Ch
2,048 Mbps
G.732
G.735
G.736
G,737
X 4
120 Ch
8,448 Mbps
G.742
G.744
G.745
X 4
480 Ch
34,368 Mbps
G.751
G.753
X 4
1920 Ch
139,264 Mbps
G.751
G.754
X 4
7680 Ch
560 Mbps
Eropa, Indonesia

➢ ITU RECOMMENDATIONS - PDH

- **G.701**
  Vocabulary of digital transmission and multiplexing, and pulse code modulation (PCM) terms
- **G.702**
  Digital hierarchy bit rates
- **G.703**
  Physical/electrical characteristics of hierarchical digital interfaces
- **G.704**
  Synchronous frame structures used at 1544, 6312, 2048, 8448 and 44 736 kbit/s hierarchical levels
- **G.705**
  Characteristics of plesiochronous digital hierarchy (PDH) equipment Functional blocks
- **G.706**
  Frame alignment and cyclic redundancy check (CRC) procedures relating to basic frame structures defined in Recommendation G.704

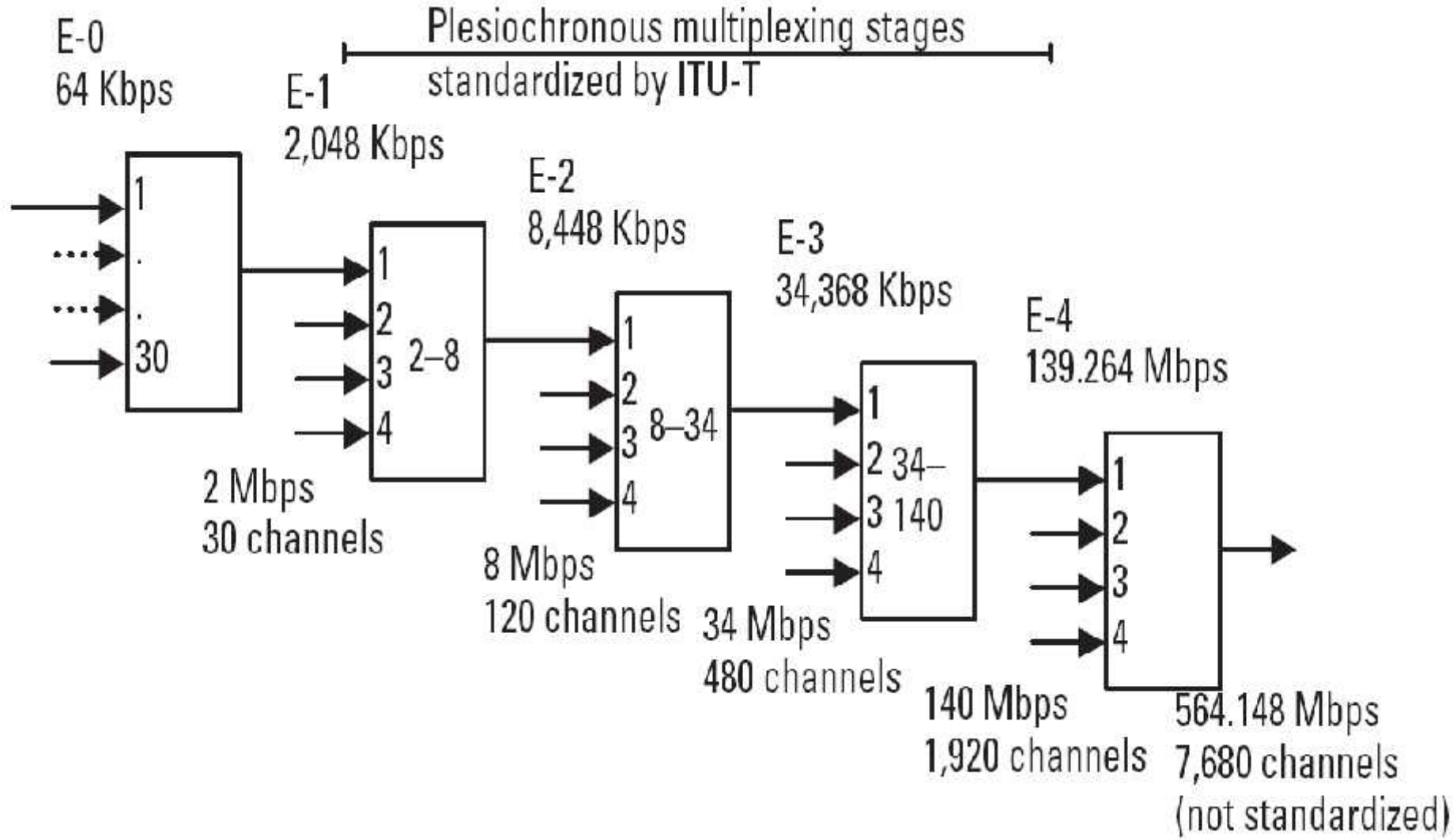
➢ Permasalahan Framing and overhead pada PDH

- In addition to locking on to bit-rate
  we need to recognize the frame structure
- We identify frames by adding **F**rame **A**lignment **S**ignal (FAS)
- The FAS is part of the *frame overhead (which also includes "C-bits", OAM, etc.)*
- Each layer in PDH hierarchy adds its own overhead

For example

- E1 – 2 overhead bytes per 32 bytes – overhead **6.25 %**
- E2 – 4 E1s = 8.192 Mbps out of 8.448Mbps
  so there is an *additional* 0.256 Mbps = 3 %
  altogether 4*30*64 kbps = 7.680 Mbps out of 8.448 Mbps
  or **9.09%** overhead

➢ Mari kita lihat kembali PDH



- Perhatikan bahwa kecepatan keluaran setiap multiplexing tingkat tinggi adalah kira-kira lebih dari 4 kali kecepatan sinyal *tributary* (bukan tepat 4 kali kecepatan sinyal *tributary*)
  - Contoh: Kecepatan E-2 adalah 8,448 Mbps (ini tidak sama dengan 4x2,048 Mbps)
  - Pada keluaran masing-masing multiplexer juga ada informasi batas frame
- Keluaran setiap level merupakan susunan *bit interleaved* dari setiap sinyal tributary
  - Artinya, keluaran setiap hirarki tersusun dari satu bit yang berasal dari tributary 1, satu bit dari tributary 2, 3 , 4, lalu dari tributary 1 lagi dst.
- Ingat: pada PDH, kecepatan masing-masing sinyal tributary boleh berbeda sedikit

- Oleh karena itu, sebelum dimultiplex, kecepatan masing-masing sinyal tributary harus disesuaikan agar ketika dimultiplex akan diperoleh kecepatan yang sesuai pada setiap tingkat
  - Penyesuaian kecepatan ini disebut *justification* atau *stuffing*
- *Justification/stuffing* dilakukan dengan cara menambahkan sejumlah bit justifikasi kepada setiap tributary
- Pada demultiplexer, bit-bit justifikasi ini dihilangkan sehingga rate tributary asli dapat kembali diperoleh
- Kondisi yang sama terjadi pada PDH versi Amerika Utara

➢ PDH overhead

digital signal	data rate (Mbps)	voice channels	overhead percentage
T1	1.544	24	0.52 %
T2	6.312	96	2.66 %
T3	44.736	672	3.86 %
T4	274.176	4032	5.88 %
E1	2.048	30	6.25 %
E2	8.448	120	9.09 %
E3	34.368	480	10.61 %
E4	139.264	1920	11.76 %

- Overhead always increases with data rate !

➢ OAM

- analog channels and 64 kbps digital channels
  do not have mechanisms to check signal validity and quality

thus

- major faults could go undetected for long periods of time
- hard to characterize and localize faults when reported
- minor defects (kerusakan) might be unnoticed indefinitely

Solution is to add mechanisms based on overhead

as PDH networks evolved (berkembang), more and more overhead was dedicated to **O**perations, **A**dministration and **M**aintenance (OAM) functions

including:

- monitoring for valid signal
- defect reporting
- alarm indication/inhibition (AIS)

➢ PDH Justification

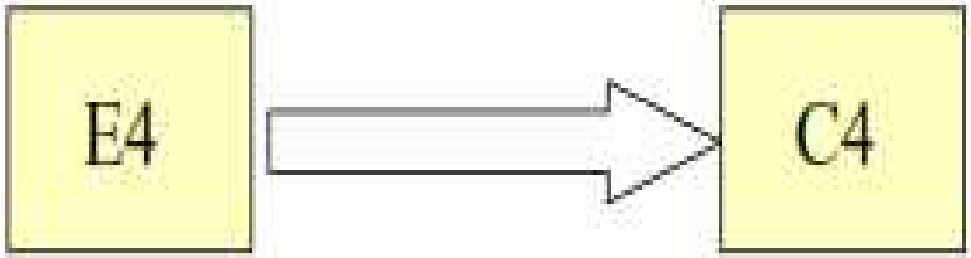
- In addition to FAS, PDH overhead includes justification control (C-bits) and justification opportunity“stuffing” (R-bits)
- Assume the tributary bitrate is B ï T

❖ Positive justification

- payload is expected at highest bitrate B+T
- if the tributary rate is actually at the maximum bitrate then all payload and R bits are filled
- if the tributary rate is lower than the maximum then sometimes there are not enough incoming bits so the R-bits are not filled and C-bits indicate this

- **Negative justification**
  - payload is expected at lowest bitrate B-T
  - if the tributary rate is actually the minimum bitrate then payload space suffices
  - if the tributary rate is higher than the minimum then sometimes there are not enough positions to Accommodate so R-bits in the overhead are used and the C-bits indicate this
- **Positive/Negative justification**
  - payload is expected at nominal bitrate B
  - positive or negative justification is applied as required

➢What is the «Justification» ?

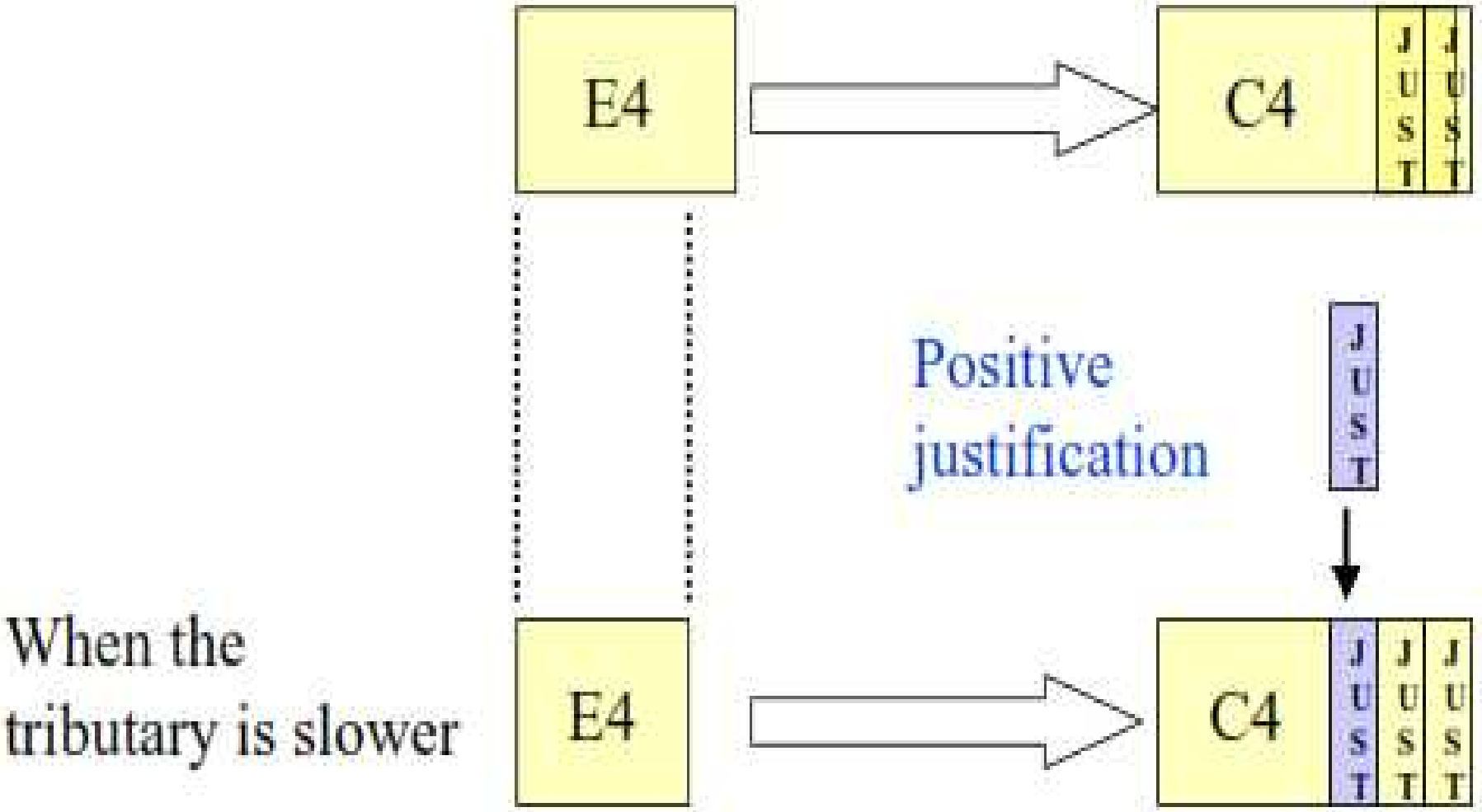


In theorie, the E4 speed should be the same than the C4 speed.

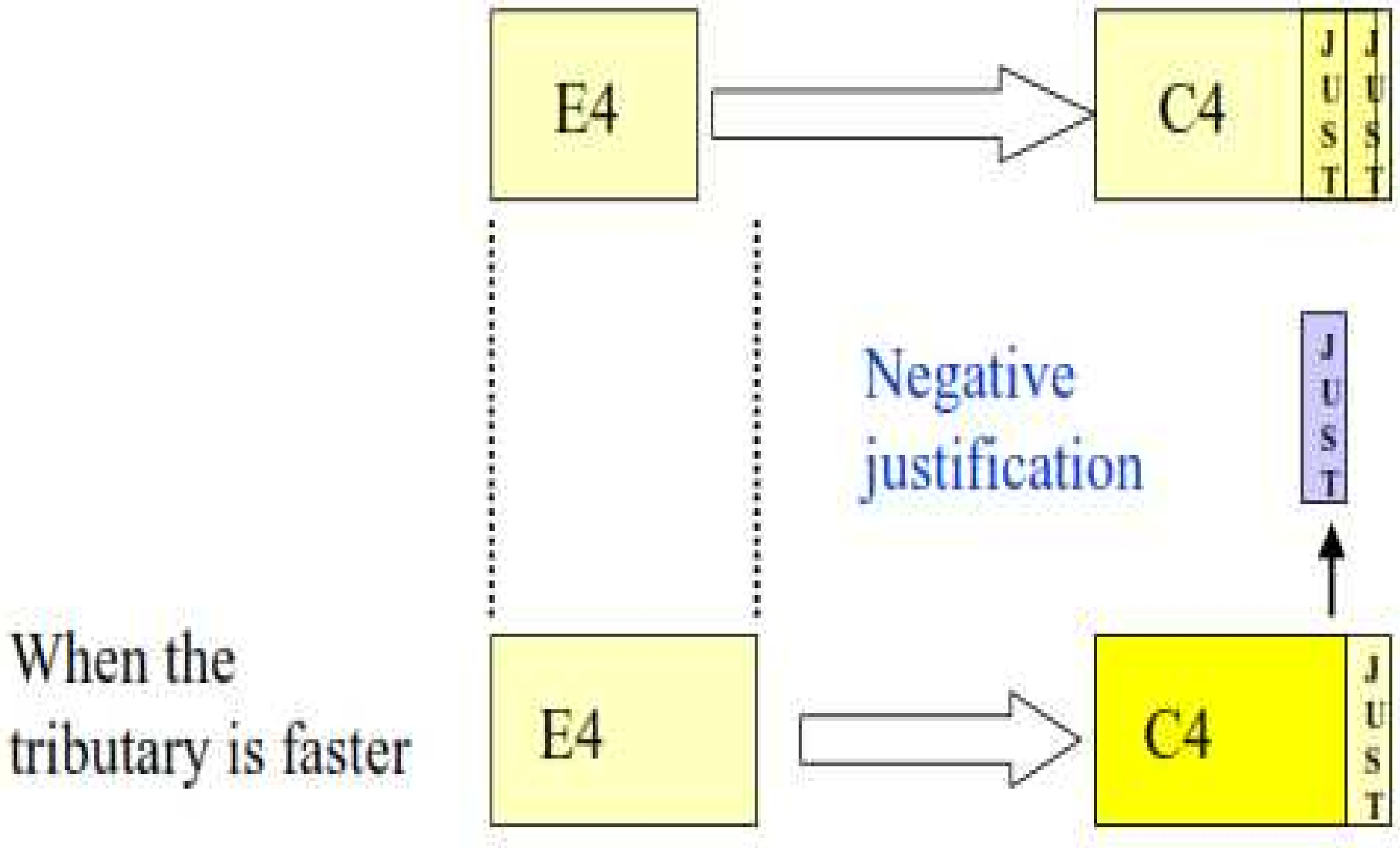
But in pratical, the E4 speed can be a little bit faster or slower than the theorical speed.

If you want to adapt the speed variation, you need a special system called « justification » each time you want to map a tributary.

What is the «J usti fication» ?



What is the «J usti fication» ?



➢ PDH limitations

Rate limitations

- Copper interfaces defined
- Need to mux/demux hierarchy of levels (hard to pull out a single timeslot)
- Overhead percentage increases with rate

- At least three different systems (Europe, NA, Japan)
  - **E** 2.048, 8.448, 34.348, 139.264
  - **T** 1.544, 3.152, 6.312, 44.736, 91.053, 274.176
  - **J** 1.544, 3.152, 6.312, 32.064, 97.728, 397.2
- **Transmisi diatas 140 Mbps tidak ada standar baku untuk setiap vendor**

So a completely new mechanism was needed

- PDH tidak begitu cocok untuk mendukung perkembangan teknik pengendalian dan pemrosesan sinyal untuk masa kini yang makin banyak dibutuhkan oleh perusahaan-perusahaan penyedia layanan telekomunikasi. Dalam PDH, sebuah peralatan transmisi tertentu umumnya hanya menangani dengan baik satu fungsi tertentu saja dalam jaringan, sementara dalam SDH, ada integrasi dari berbagai tipe peralatan yang berbeda-beda yang mampu memberikan kebebasan baru dalam perancangan jaringan.

➢ Pengertian SDH

- ***Pengertian SDH menurut Rec-707 ITU-T adalah*** “teknologi transport digital yang tersusun secara hierarki untuk multipleksing dan membawa sinyal informasi (payload) melalui media transmisi fisik. Media transmisi ini pada umumnya serat optik, selain itu memungkinkan juga digunakan link radio teresterial maupun link satelit. “

➢ Perbandingan Teknologi SDH dan PDH

PDH

1. **Jenis sinyal plesinkron, yaitu adanya pergeseran clock.**
2. **Bit rate dasar sebesar 1.544 Kb/s (PCM-24) biasa disebut sinyal T1, atau 2.048 Kb/s (PCM-30) biasa disebut sinyal E1.**
3. **Teknik multipleks bit per bit.**
4. **Penyelarasan terhadap bit rate dari frame dilakukan dengan metode jastifikasi positif.**
5. **Penyelarasan phase menggunakan buffer memory.**
6. **Setiap tahapan (orde) multipleks memiliki struktur frame yang berbeda.**
7. **Pengaksesan sinyal selalu melalui prosedur bertingkat.**

SDH

1. **Jenis sinyal sinkron, yaitu dengan adanya clock acuan.**
2. **Bit rate dasar sebesar 155.52 Mb/s (STM-1).**
3. **Teknik multipleks byte per byte.**
4. **Sinkronisasi menggunakan metode jastifikasi positif, negatif, dan nol.**
5. **Penyelarasan phase menggunakan teknik pointer.**
6. **Setiap tahapan multipleks memiliki struktur frame yang identik.**
7. **Memungkinkan pengaksesan sinyal kecepatan rendah secara individu .**

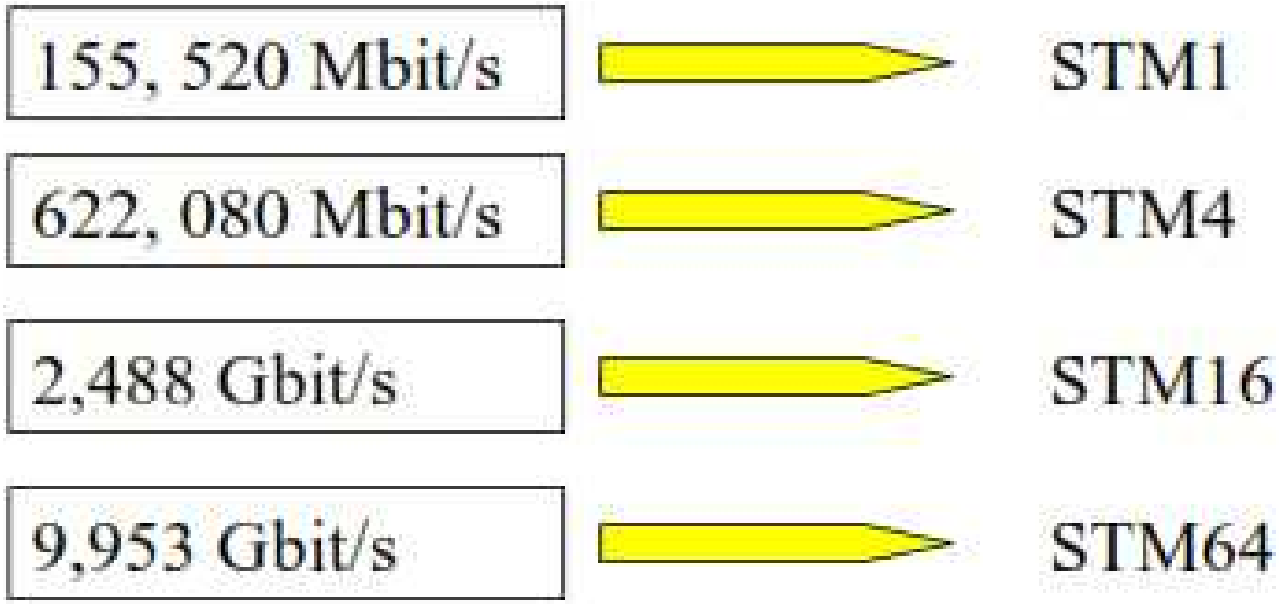
SDH Advantages versus PDH

- SDH is based on the principal of direct synchronous multiplexing.
- Essentially, separate, slower signals can be multiplexed directly onto higher speed SDH signals without intermediate stages of multiplexing.
- SDH is more flexible than PDH and provides advanced network management and maintenance features.
- Can be used in the three traditional telecommunications areas: long-haul networks, local networks and loop carriers. It can also be used to carry CATV video traffic.

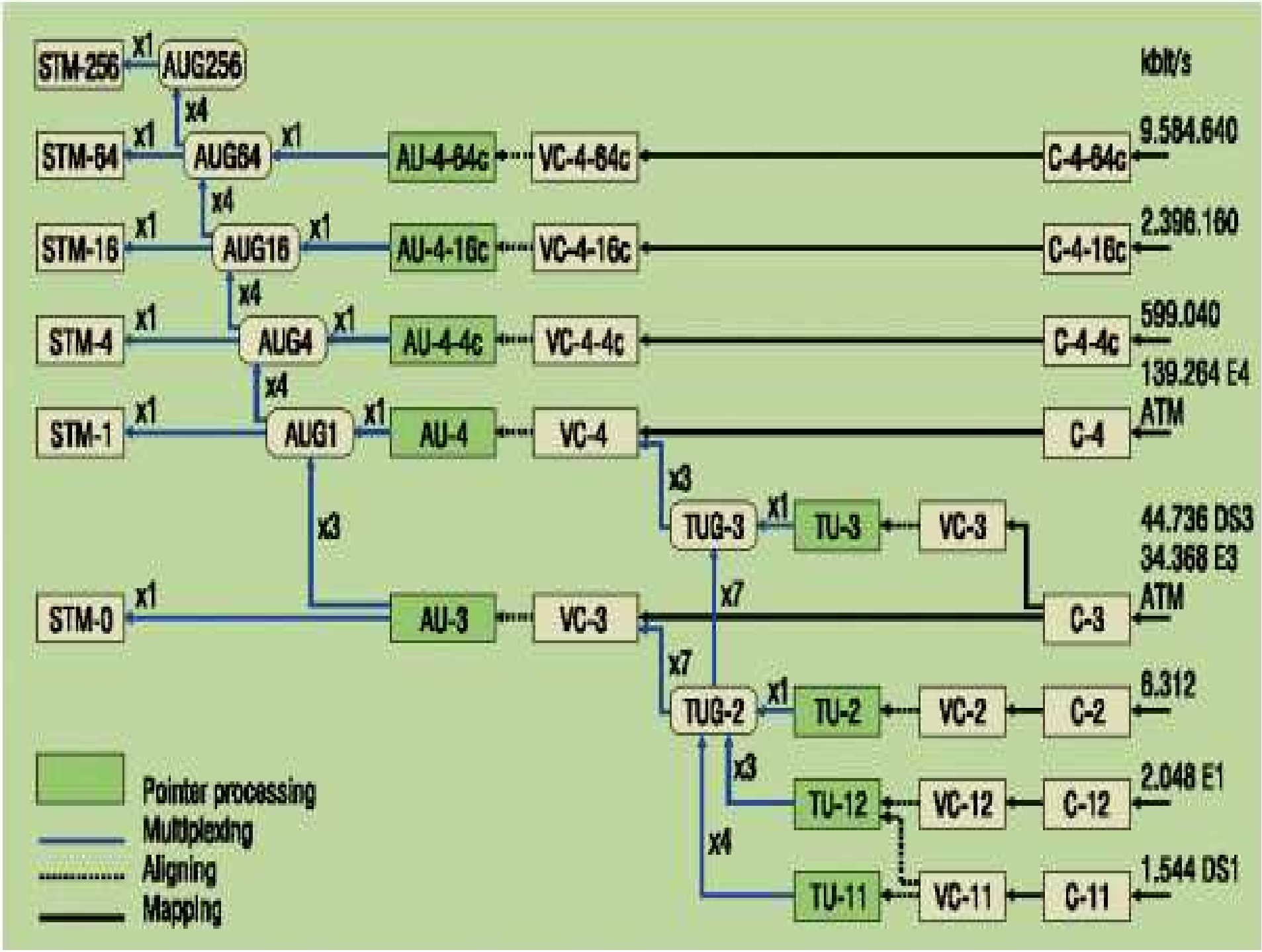
Synchronous Digital Hierarchy (SDH)

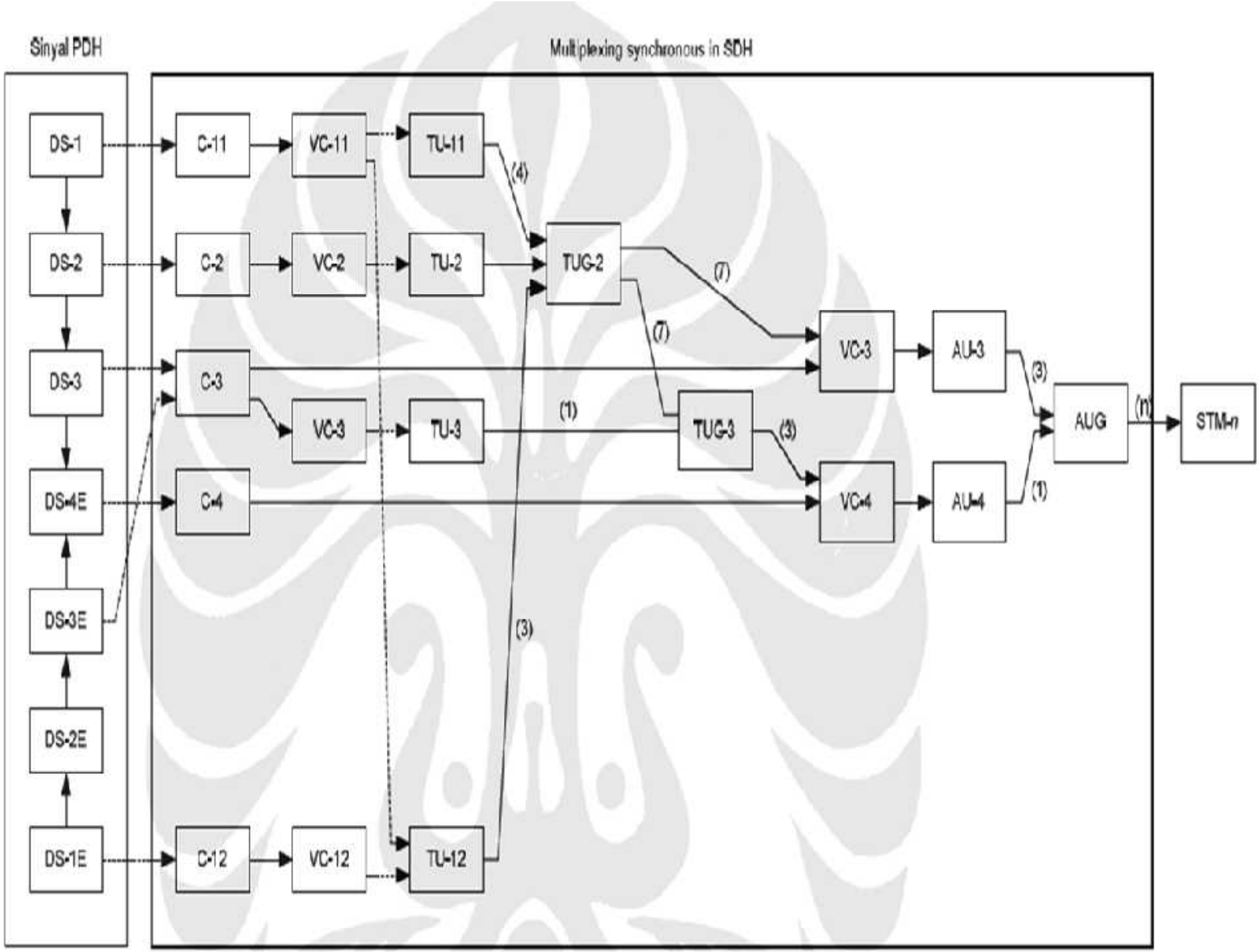
- Bit Rates

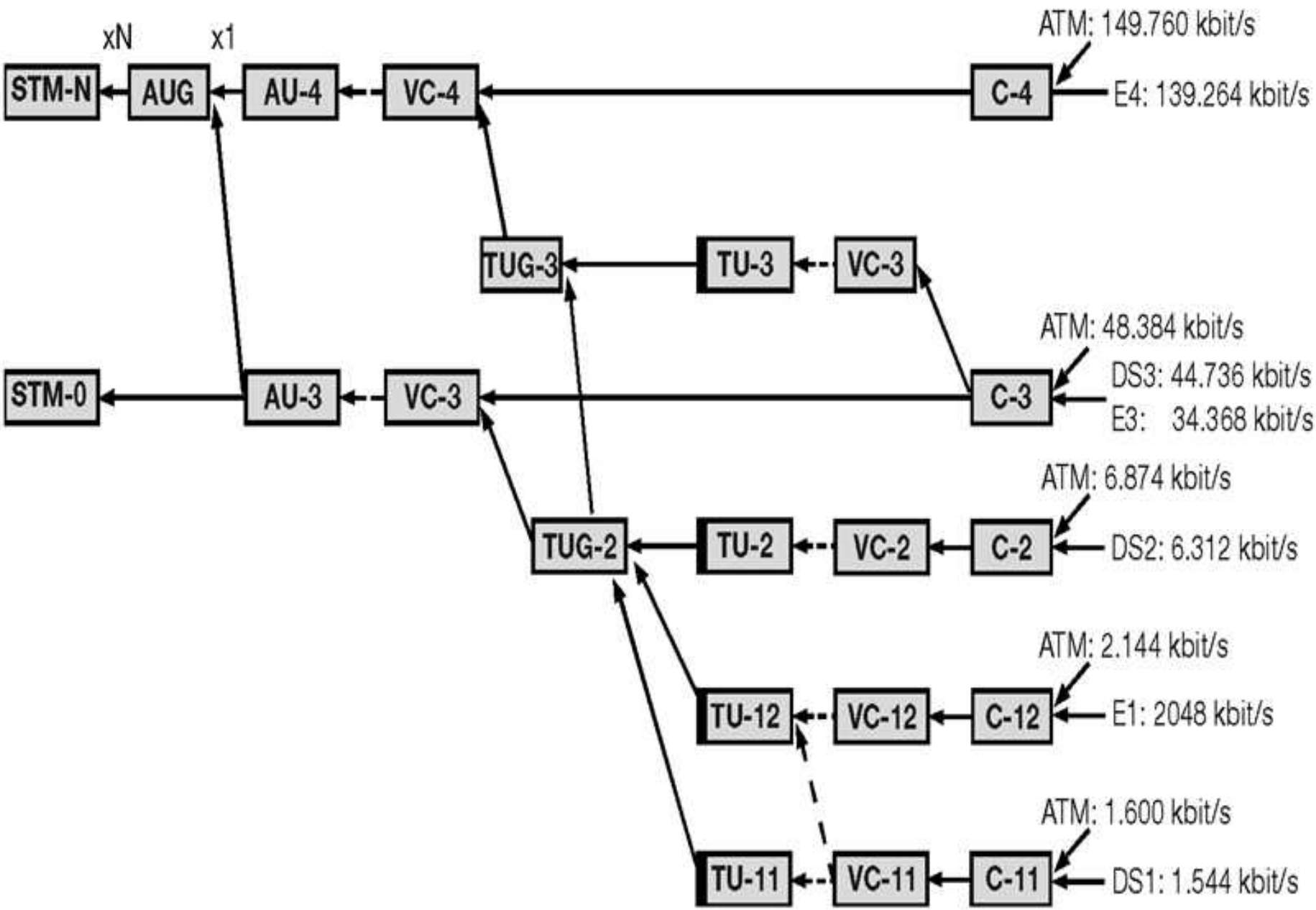
International organization defined standardized bit rates :



Dan seterusnya


STM-256
x1
AUG256
kbit/s
x4
STM-64
x1
AUG64
x1
AU-4-64c
VC-4-64c
C-4-64c
9.584.640
x4
STM-16
x1
AUG16
x1
AU-4-16c
VC-4-16c
C-4-16c
2.396.160
x4
STM-4
x1
AUG4
x1
AU-4-4c
VC-4-4c
C-4-4c
599.040
x4
139.264 E4
ATM
STM-1
x1
AUG1
x1
AU-4
VC-4
C-4
x3
TUG-3
x1
TU-3
VC-3
44.736 DS3
34.368 E3
ATM
x3
x7
STM-0
x1
AU-3
VC-3
C-3
x7
TUG-2
x1
TU-2
VC-2
C-2
6.312
x3
TU-12
VC-12
C-12
2.048 E1
x4
TU-11
VC-11
C-11
1.544 DS1
Pointer processing
Multiplexing
Aligning
Mapping


Sinyal PDH
Multiplexing synchronous in SDH
DS-1
DS-2
DS-3
DS-4E
DS-3E
DS-2E
DS-1E
C-11
C-2
C-3
C-4
C-12
VC-11
VC-2
VC-3
VC-12
TU-11
TU-2
TU-3
TU-12
TUG-2
TUG-3
VC-3
VC-4
AU-3
AU-4
AUG
STM-n
(4)
(7)
(7)
(1)
(3)
(3)
(3)
(1)
(n)


xN
x1
STM-N
AUG
AU-4
VC-4
C-4
ATM: 149.760 kbit/s
E4: 139.264 kbit/s
TUG-3
TU-3
VC-3
ATM: 48.384 kbit/s
DS3: 44.736 kbit/s
E3: 34.368 kbit/s
STM-0
AU-3
VC-3
C-3
ATM: 6.874 kbit/s
TUG-2
TU-2
VC-2
C-2
DS2: 6.312 kbit/s
ATM: 2.144 kbit/s
TU-12
VC-12
C-12
E1: 2048 kbit/s
ATM: 1.600 kbit/s
TU-11
VC-11
C-11
DS1: 1.544 kbit/s

Sekian Dulu kita

sambung kuliah

minggu depan